

BIRSK, Russian Federation

WMO: 286210

Lat: **55.4050N**Lon: **55.5517E**Elev: **611**StdP: **14.37**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-11.9	-24.4	1.5	-18.3	-17.6	2.2	-11.2	26.4	24.9	19.8	18.7	4.4	20	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	86.4	68.4	83.6	67.1	80.5	65.6	70.9	82.2	69.1	80.4	67.3	77.6	6.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.9	101.2	77.2	65.0	94.5	75.0	63.3	88.9	73.5	35.1	82.3	33.6	80.3	32.1	77.7	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.0	14.4	12.7	-22.5	90.7	6.8	3.5	-27.4	93.3	-31.4	95.3	-35.2	97.3	-40.2	99.9		
(5)			-22.4	72.9	6.8	2.7	-27.3	74.8	-31.3	76.4	-35.0	77.9	-40.0	79.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	10.3	12.6	24.8	41.7	56.5	64.0	67.9	64.3	53.5	40.5	25.3	14.5
	DBStd	22.85	12.06	11.71	8.30	10.35	9.71	8.22	6.99	7.70	8.49	8.32	10.65	12.41
	HDD50	5619	1232	1047	781	295	47	5	0	1	58	313	740	1100
	HDD65	9552	1697	1467	1246	701	295	116	52	108	357	758	1190	1565
	CDD50	1904	0	0	0	45	249	427	555	446	162	20	0	0
	CDD65	361	0	0	0	2	31	87	142	88	11	0	0	0
	CDH74	2756	0	0	0	11	249	667	1107	658	64	0	0	0
	CDH80	754	0	0	0	1	43	179	320	202	9	0	0	0
	WSAvg	6.0	6.5	6.5	6.7	6.1	6.2	5.7	5.0	5.0	5.4	6.2	6.2	6.4
	PrecAvg	21.9	1.5	1.3	1.3	1.1	1.7	2.6	2.1	2.0	1.9	2.4	2.0	1.9
(15) Precipitation	PrecMax	29.2	2.9	2.7	2.4	2.7	3.1	5.4	5.0	4.6	4.4	4.8	4.6	3.7
	PrecMin	14.7	0.2	0.3	0.1	0.0	0.2	0.3	0.6	0.3	0.2	0.4	0.3	0.4
	PrecStd	4.1	0.7	0.6	0.7	0.7	0.9	1.4	1.3	1.0	1.0	1.3	1.1	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	35.4	46.6	74.9	84.6	89.3	90.5	90.7	81.0	65.9	49.2	35.4
	0.4%	MCWB	32.5	32.5	39.4	57.2	63.2	68.7	69.0	68.8	60.7	53.2	46.0	32.8
	2%	DB	31.6	33.1	41.4	68.4	80.5	84.9	86.3	85.4	75.1	60.8	43.9	33.9
	2%	MCWB	30.4	31.4	36.0	53.6	61.8	67.5	69.4	68.1	59.2	50.2	41.6	32.5
	5%	DB	29.5	31.1	37.9	63.4	76.5	81.2	83.7	80.9	70.3	56.3	40.4	32.4
	5%	MCWB	28.3	29.7	33.7	50.9	59.8	65.9	68.5	65.9	58.1	48.6	38.3	31.2
	10%	DB	26.8	28.3	35.3	57.8	72.3	77.6	80.6	76.9	66.2	52.4	37.2	30.2
	10%	MCWB	25.5	26.7	32.2	47.7	57.3	64.4	67.2	64.6	56.2	46.8	35.2	28.9
	0.4%	WB	33.1	33.2	40.2	58.7	66.4	73.4	73.8	72.7	64.4	56.3	46.3	34.1
	0.4%	MCDB	33.9	34.6	45.3	73.4	80.7	86.1	84.3	83.0	73.7	61.6	48.9	34.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	30.7	31.6	36.5	54.5	63.0	69.4	71.5	69.2	61.3	52.4	41.9	32.7
	2%	MCDB	31.5	32.9	40.5	66.8	77.0	80.9	82.6	81.9	70.1	58.0	43.6	33.5
	5%	WB	28.4	29.8	34.3	51.5	60.8	67.2	69.9	67.2	59.1	49.4	38.5	31.2
	5%	MCDB	29.3	31.1	37.0	62.7	74.3	77.8	81.3	79.0	68.3	54.9	40.1	32.3
	10%	WB	25.6	26.8	32.5	48.0	58.6	65.4	68.3	65.2	57.1	47.1	35.5	29.1
	10%	MCDB	26.8	28.1	34.9	57.5	70.6	75.6	78.1	75.0	64.9	51.7	36.7	30.3
(35) Mean Daily Temperature Range		MDBR	9.5	10.6	11.1	13.6	16.5	15.8	15.9	15.0	13.1	9.4	7.4	8.8
	5% DB	MCDBR	9.0	9.0	11.9	19.6	21.3	20.5	19.8	20.0	18.8	15.9	8.1	8.1
	5% DB	MCWBR	9.0	8.3	8.2	10.8	10.1	9.1	7.9	8.2	9.1	9.0	7.0	8.0
	5% WB	MCDBR	9.5	8.7	9.9	19.3	19.8	18.5	17.9	18.6	16.8	14.0	7.9	7.9
	5% WB	MCWBR	9.6	8.4	7.3	11.4	10.4	9.6	8.4	8.7	9.2	9.5	7.4	7.8
(40) Clear-Sky Solar Irradiance	taub		0.294	0.318	0.370	0.408	0.396	0.390	0.408	0.427	0.379	0.354	0.315	0.293
	taud		2.031	2.025	1.950	2.081	2.268	2.341	2.296	2.224	2.326	2.274	2.158	1.999
	Ebn at Noon		184	229	244	255	268	271	263	248	245	218	179	150
	Edh at Noon		22	32	44	44	39	37	38	38	30	25	19	18
(44) All-Sky Solar Radiation	RadAvg		252	570	1031	1393	1709	1848	1833	1427	893	399	182	145
	RadStd		29	58	93	148	153	173	179	133	87	87	40	32

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page